Coordination, Measurement, and Iteration (vF and Chang) April 8, 2026

- Proposal: an account of measurement is an essential part of how scientific theories "get their content," a response to Duhem's question regarding how we interpret and apply symbolic representations / mathematical formalism. But how does this work in detail? What is the proper role and status of conventions, choices, historical development of the field, ...?
- How do vF and Chang define measurement and its role, and how do their accounts differ? What are their criteria of adequacy for an account of measurement, and what philosophical problems does it solve?
- What are the consequences of this account? What does a successful measurement practice establish – in what sense should we treat measurable properties as "objective" or "real," as opposed to merely empirically adequate?

Conventionalism and the Problem of Coordination

Debates regarding conventionalism in early 20th century (see also, e.g., [Friedman, 1999, DiSalle, 2002]):

- Discovery of variety of logically consistent geometries. What is the status of geometrical knowledge, and what is the geometry of physical space? Conventionalism as alternative to two other unappealing views: geometry fixed *a priori*, or determined empirically. (Reject one way of assigning empirical content: "impose a parallel vocabulary and declare victory" (vF, p. 137), as begging the question.) Significant influence on logical positivists; later debates with Grünbaum et al.
- Coordinative definition: non-empirical law, establishes a connection between elements of the mathematical formalism and physical systems. Poincaré, Reichenbach, ...: *stipulative* or *conventional* character of these definitions; several distinct possibilities, choice among them to be based on simplicity (or other pragmatic grounds). Further, these definitions are not fixed *a priori* and can be revised / change over time.
- Poincaré: measurement of time, limitations of various candidate "clocks" (pendulums, rotation of the earth, ephemeris time ...)
- Reichenbach: determination of the physical geometry of a surface presupposes account of measurement, e.g. stipulation that measuring rods retain their length as they move from one region to another.

vF's Reformulation

Earlier formulations of problem of coordination formulated as if from a "view from nowhere"; historical cases motivate articulation of two points of view that we should combine in a "synoptic vision":

- View "from within": Measurement practices always begin *in media res*, no "neutral starting point," and aim to extend existing measurements to introduce new quantities. Epistemological holism of any theory that treats its own measurement devices (as with physics): there are choices involved in interplay between theory and measurement practices, as new quantities are introduced, but they are not *conventional* – rather, they are constrained by coherence conditions. (We cannot single out one choice as *conventional* and others as *compelled*; contra Poincaré, choices are not dictated solely by simplicity.)

- View "from above": if stability has been achieved, we can (mostly) ignore theory-dependence. (Stability is an empirical fact: the success in using a theory with some set of quantities in describing some domain of phenomena.) We have accepted a theoretical description that applies to target systems and also underwrites an account of how we can measure their properties (without presupposing "an impossible god-like view," p. 139).

- Questions: how should we describe stabilization (in a way that is "legitimate from an empiricist point of view"), and what follows from it? What is the nature of the coherence conditions? What is required, if anything, beyond coherence? To what extent can we minimize contingency — that is, is there a sense in which different choices nonetheless lead to the "same outcome"?

Measurement as Representation

Dual aspects of measurement (vF, Chapters 6–7, see also [Van Fraassen, 2009, Van Fraassen, 2012]):

- "Physical correlates" of measurement interaction and outcome: abstract characterization of minimal pre-conditions for measurement. (Contrasts with other forms of representation: physical interactions of measurement device and target system have particular weight; coherence conditions.)

- Propositional / informational content: outcome of measurement as having particular representational role, "locating attributes of the target system in a logical space." (Consideration of different types of "logical spaces," resolve apparent tension with perspectival aspect of measurement.)

General Theory of Measurement

Simple account (note that everything is described "from above," relative to a given background theory):

- Target system T, with properties $\{P_1, P_2, P_3 ...\}$, each with two values ("yes" and "no").

  State space of T: state of the system gives full specification of the values of all properties. State space Γ represents the collection of all possible states T could be in, with a specific state given by $p \in \Gamma$. (Γ can come equipped with different mathematical structures.)

- Measuring apparatus M, with "pointer-variable" V, with two values (V_1, V_2).

- Measurement interaction for property P_i: dynamical evolution on the combined $T + M$ system, such that the pointer variables (V_1, V_2) reliably indicate the value of P_i for all possible states (i.e., V_1 iff P_i is "yes" for the given state, and V_2 iff P_i is "no")

Towards a more sophisticated account (at the level of physical correlates):

- Generalize from properties that admit only two values to richer value spaces (e.g. topological space, $\mathbb{R}$, ...).

- Introduce probabilities in two senses: (i) uncertainty regarding precise $p \in \Gamma$; (ii) stochastic measurement interaction (noise, instrument limitations).

- Simple account includes commitment to measurement as "revelation" of pre-existing definite value. vF's criterion (p. 152) compatible with dropping both value-definiteness and veracity in light of QM. (Minimalist, theory-neutral approach.)

- What is required for successful measurement? "Coherence": agreement of repeated measurements of the same type, given that we can couple the same target system T to M' of the same type.

  We should augment vF's account to acknowledge three different considerations related to "coherence" (following Smith and Seth): (1) stability – repeated measurements of the same type leading to compatible results; (2) convergence – agreement among different types of measurement devices that target the same underlying property P_i; (3) amenability to increasing precision – the ability to maintain stability and convergence while increasing precision.

Information

Grant the physical account of M, T and their interaction. Further questions before we can treat the pointer-variable as representing a "measurement outcome":

- *Calibration*: what does an outcome mean? Introduce a scale, and a mapping from V onto this scale.
- *Appropriate Structure*: how much can we infer from numerical assignments? (Is $V = 6$ "twice as much as" $V = 3$? Is the difference between $V = 6$ and $V = 5$ the same as between $V = 3$ and $V = 2$?)

Representational theory of measurement: how should we characterize the structure of the properties attributed to the target systems, and preserve this structure as it is mapped into value assignments? Consider what properties are invariant under transformations of a measurement scale; require that these match the "meaningful structural features" of P.

Towards a more sophisticated account:

- *Logical space*: the appropriate structures need not be limited to $\mathbb{R}$; more generally allow mapping to any structured space that represents the range of possible values.
- *Approximate Results*: measurements yield intervals of values, which changes the character of the required mathematical structures.
- *Data and Surface Models*: shift from single measurements to repeated measurements; construct summaries (relative frequency, mean, standard deviation) and "surface models" (continuous functions fit to discrete data). Typically data or surface models (rather than individual outcomes) are checked for compatibility with theory.
- *Locating vs. Perspectival*: *pace* Poincaré, vF argues that even measurements that start out as inherently perspectival can lead to characterizations of invariant quantities (e.g., from the captain's observations of the North Star to the ship's location on a grid).

Chang: The History of Thermometry

Chang's *Inventing Temperature* provides a detailed historical case study that bears directly on the questions raised by vF's account. Thermometry developed largely *without* the aid of a mature background theory, making it especially revealing regarding the role of coherence, circularity, and iterative refinement.

Stages in the development of temperature measurement (Chapter 1):

- *Stage 1*: bodily sensations of heat and cold (qualitative, unreliable, but providing an initial ordering)
- *Stage 2*: "thermoscope" – an instrument that enables comparisons and ordering (based on expansion of fluids), without calibration or numerical assignments

- *Stage 3*: fixed points (blood heat; freezing and boiling of water) evaluated by thermoscopes, then used to establish numerical scales and calibrate instruments

Note that each stage builds on the preceding one ("principle of respect"), and the transition from one stage to the next requires substantive steps (e.g., establishing the fixity of fixed points).

The Problem of Nomic Measurement (Chapter 2): We want to measure quantity X (temperature), but X is not directly accessible; we infer it from an observable proxy Y (volume of thermometric fluid) via a functional relationship $X = f(Y)$. But how can we determine the form of f without already having access to X? The usual practice of dividing the scale linearly between fixed points *assumes* f is linear – an assumption that cannot be tested without a prior, independent standard.

- *Disagreement among thermometers*: mercury and alcohol thermometers, calibrated at the same fixed points, give *different readings* at intermediate temperatures – a significant, not subtle, discrepancy. Which gives the "real temperature"?

- *Method of mixtures* (De Luc): mix boiling and freezing water in known proportions; the "real temperature" should be determined by the ratio. Presupposes that the specific heat of water does not vary with temperature – an assumption undermined by caloric theory and later confirmed as incorrect.

- *Regnault's comparability tests*: demanding comparisons of the same type of thermometer (stability) and direct comparisons of different types (convergence). Conclusion: there is no ideal thermometric fluid, but air thermometers are the best available.

Role of theory: (i) measurement scales and fixed points established without the relevant modern theory (thermodynamics, conservation of energy); (ii) thermometry survived the rise and fall of the caloric theory; (iii) yet, the lack of background theory also limited what could be achieved (Regnault's pessimism about measuring "absolute quantities of heat").

Coherence and Epistemic Iteration

Circularity and coherentism (Chapter 5): The problem of nomic measurement is an instance of a pervasive circularity in empirical inquiry: we need measurement to test theories, but measurement presupposes (at least some) theoretical commitments. Chang argues that empiricist foundationalism cannot solve this problem; the only viable response is a form of *coherentism*.

Epistemic iteration as a method for making coherentism progressive:

- *Successive stages*: each stage of knowledge builds on the preceding one, incorporating a "principle of respect" for what has been established, while seeking to extend and correct it.

- *Enrichment*: enhancement of epistemic virtues (precision, scope, explanatory power, ...) through successive iterations

- *Self-correction*: the content of the initial system is altered based on discoveries made within the framework it provides

- *Risk of self-destruction*: since the affirmed system is subject to modification, there is a possibility that the validity of the inquiry itself will be jeopardized. Not every iteration succeeds; the process can be self-undermining rather than self-correcting.

Comparability and the Principle of Single Value: Regnault's criterion of comparability (agreement among thermometers of different construction) provides a theoretically neutral test of reliability. The underlying ontological commitment is minimal: the "principle of single value" – a determinable physical quantity can have only one definite value in a given situation. This is not a logical principle but a physical one, reflecting our understanding of temperature (and other physical quantities).

Minimalism against Duhemian holism: Chang emphasizes Regnault's success in minimizing auxiliary assumptions. Rather than relying on theoretical derivations from the caloric theory (or any other), Regnault tested thermometers against each other, reducing the theoretical burden as far as possible. This is an empiricist strategy that attempts to break free of holistic entanglement by finding comparisons that depend on very few background assumptions.

Pluralism: Chang's account is pluralistic in several senses: (i) the relevant epistemic virtues are plural (precision, scope, explanatory power, etc.), and enhancing one may come at the cost of another; (ii) the choice of which existing system to affirm as a starting point is not uniquely determined; (iii) the direction of iterative development from any starting point remains open.

Comparing vF and Chang

- Both emphasize that measurement practices begin *in media res*, without neutral foundations; both reject the idea that the problem of coordination can be solved by stipulative definition alone.
- vF treats the problem of coordination as resolved when measurement practices "stabilize" within a theoretical framework (view "from above"); his account of what stabilization requires – and what epistemic status it confers – remains somewhat underdeveloped. Chang offers a detailed historical picture of how stabilization has been achieved, in one case, through iterative refinement and comparability testing.
- vF emphasizes *coherence* but from within an empiricist framework that resists attributing more than empirical adequacy to the result. Chang is more willing to treat comparability as establishing "observability" – i.e., to regard successful measurement as a genuine epistemic achievement, not merely a pragmatic or conventional one. (Chang: "observability is an achievement.")
- Regarding the role of theory: vF's account is heavily theory-dependent (measurement is described relative to a background theory). Chang's case study highlights a context where measurement develops with minimal understanding: theory relevant to reliability of measurement methods, but with little theoretical understanding of the nature of heat / thermodynamics.

Questions for Discussion

1. Chang argues that Regnault's comparability criterion provides a "non-theoretical" standard of reliability. Is this right? What background assumptions does comparability still presuppose (e.g., the principle of single value, the stability of fixed points)? Is it possible to have a theory-neutral assessment of the success of a measurement practice based on things like stability, convergence, and amenability to increased precision?
2. How should we understand the epistemic status of a measurement practice that has been "stabilized" (vF) or that has achieved "comparability" (Chang)? Does successful measurement establish that the measured quantity is *real*, or merely that our measurement practices are internally coherent? What would it take to distinguish these?

3. Chang's account of epistemic iteration involves a "principle of respect" for earlier stages of inquiry. How should we understand this principle? Under what circumstances is it legitimate to *violate* it – i.e., to abandon rather than refine an earlier stage? Are there cases where iterative development continues to provide further evidence in favor of the basic framework of inquiry? If so, what are the characteristic features of this form of progressive iteration?

4. Both vF and Chang appeal to forms of coherence as a constraint on measurement. But coherence among *what*? Is it enough for different instruments to agree (convergence), or do we also need the measurements to cohere with a broader theoretical framework? How do we handle cases where instruments agree but the theoretical account of *why* they agree is unavailable or contested?

5. Chang emphasizes that thermometry developed largely without the aid of mature theory, and treats this as a virtue of the case study. But is this feature typical or exceptional? In cases where measurement is more heavily theory-mediated (e.g., particle physics, cosmology), does Chang's iterative-coherentist picture still apply? (Can we say something stronger about what is established in such cases, or are there new problems?)

References

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